

DELHI PUBLIC SCHOOL BANGALORE NORTH
REMEDIAL WORKSHEET

CLASS: VIII

SUB: MATHEMATICS

TOPIC: FACTORIZATION

1. Factorize using identities

a) $81p^2 q^2 - 49$

b) $16a^2 - 25/4a^2$

2. Factorize using identities

a) $h^2 - 13h - 30$

b) $x^2 - 8xy - 48y^2$

c) $p^2 + p - 72$

3. Factorize and divide

a) $(p^2 + 11p + 28) / (p + 4)$

b) $4yz(z^2 + 6z - 16) / 2y(z + 8)$

4. Divide using long division

a) $6x^2 + 7x - 20$ by $2x + 5$

b) $3x^3 + 4x^2 + 5x + 18$ by $x + 2$

5. Solve using identity

a) $51^2 - 49^2$

b) $(1.02)^2 - (0.98)^2$

6. Solve:

$(x^4 - 1) / (x - 1)$

7. Verify whether the following equations are correct. Rewrite correctly.

(i) $(a + 6)^2 = a^2 + 12a + 36$

(ii) $(2a)^2 + 5a = 4a + 5a$

8. Solve for:

$(4x^2 - 100) / 6(x + 5)$

9. Factorize each of the following by regrouping:

(i) $x^2 + xy + 9x + 9y$

(ii) $6xy - 4y + 6 - 9x$

10. State True(T) Or False(F):

1. $(2x - 5) / 2x = -5$

2. $3(y - 2) = 3y - 2$

3. $4x + 3y = 7xy$

4. $(3x)^2 + 4(3x) + 5 = 3x^2 + 12x + 5$

5. $a(5a + 2) = 5a^2 + 2a$